

The empirical recurrence for OEIS A259950, recovered from its tail

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Abstract

OEIS A259950 records “Empirical recurrence of order 67” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 92$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{67} = -576 \neq 0$, so no further tail satisfies anything shorter; any order-67 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A259950 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000000 00000001 or 00000101.”. It has offset 1 and begins

1825, 11933, 115410, 642779, 3451108, 23837974, 159480543, 966996539, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:56:40, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 92$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 92$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 67: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 67.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 184$ exact terms from $n = 5$ on, it returns order 67 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	2	c_{18}	29248802	c_{35}	448796819	c_{52}	−53692582
c_2	9	c_{19}	10021448	c_{36}	−60969386	c_{53}	15823620
c_3	67	c_{20}	109356222	c_{37}	3226178298	c_{54}	31192238
c_4	443	c_{21}	65099487	c_{38}	−2048783936	c_{55}	−11917879
c_5	−307	c_{22}	−220615086	c_{39}	−347733932	c_{56}	2427096
c_6	−1274	c_{23}	−55310407	c_{40}	1469689467	c_{57}	2015639
c_7	−5256	c_{24}	−430569819	c_{41}	−2611754597	c_{58}	−2626707
c_8	−37959	c_{25}	−344122600	c_{42}	397139602	c_{59}	885163
c_9	−881	c_{26}	872351353	c_{43}	180744358	c_{60}	113119
c_{10}	73368	c_{27}	153116300	c_{44}	−1418677397	c_{61}	−263321
c_{11}	118194	c_{28}	975324980	c_{45}	1115603576	c_{62}	163455
c_{12}	1178092	c_{29}	1168802272	c_{46}	382620969	c_{63}	−7906
c_{13}	429326	c_{30}	−2030840883	c_{47}	−120426136	c_{64}	−31058
c_{14}	−1989139	c_{31}	−299446664	c_{48}	465992872	c_{65}	5618
c_{15}	−1177639	c_{32}	−1062692574	c_{49}	−217999307	c_{66}	−904
c_{16}	−16356702	c_{33}	−2483668949	c_{50}	−202618732	c_{67}	−576
c_{17}	−8521878	c_{34}	2829884672	c_{51}	47059157		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A259950 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{67} - c_1 x^{67-1} - \dots - c_{67}$. Its constant term is $-c_{67} = 576$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 67 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 67, so $\deg q = 67$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 67, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 67, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -576 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-67 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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